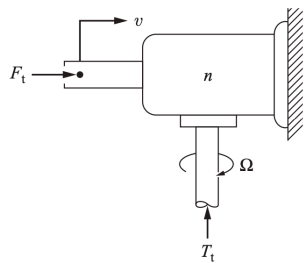


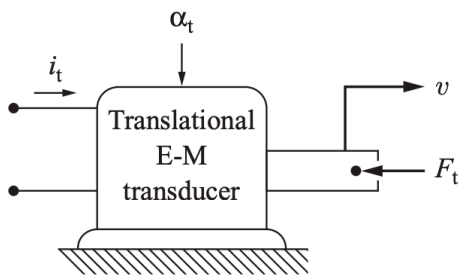
Spring Opposes Displacement <u>Translational</u> $F_s = kx(t)$ <u>Rotational</u> $T_s = \mathcal{K}\theta(t)$	Damper Opposes Velocity <u>Translational</u> $F_d = bv(t) = b \frac{dx(t)}{dt}$ <u>Rotational</u> $T_d = B\Omega(t) = \frac{d\theta(t)}{dt}$	Mass <u>Translational</u> $\Sigma F = ma = m \frac{v(t)}{dt} = m \frac{d^2x(t)}{dt^2}$ <u>Rotational</u> $\Sigma T = J\alpha = \frac{Jd\Omega(t)}{dt} = J \frac{d^2\theta(t)}{dt^2}$	
same direction different direction $\Omega_{out}(t) = \frac{1}{n} \Omega_{in}(t) \quad T_{out}(t) = nT_{in}(t)$ $n = \frac{\Omega_1}{\Omega_2} = \frac{T_2}{T_1} \quad n = \frac{n_2}{n_1} = \frac{r_2}{r_1}$	<u>Capacitor</u>: $i_c(t) = C \frac{dv_c(t)}{dt}$ <u>Inductor</u>: $V_l = L \frac{di_l(t)}{dt}$ <u>Resistor</u>: $V(t) = i(t)R$	Op Amp Characteristics $V_1 = V_2$ $i_1 = i_2 = 0A$ $-V_s \leq V_0 \leq V_s$	Voltage Follower $V_{out} = V_{in}$
Inverting Amplifier $V_{out} = -\frac{R_2}{R_1} V_{in}$	Non-Inverting Amplifier $V_{out} = \left(1 + \frac{R_2}{R_1}\right) V_{in}$	Summing Amplifier $V_{out} = -(V_1 + V_2)$	Differential Amplifier $V_{out} = \frac{R_3}{R_1} (V_2 - V_1)$
Fluid Capacitor $Q_c = C_f \frac{d(P_1 - P_r)}{dt} = C_f \frac{dP_{1f}}{dt}$	Fluid Inertor $P_{12} = I \frac{dQ_1}{dt} \quad I = \left(\frac{\rho}{A}\right) L$	Fluid Resistor $P_{12} = R_f Q_R$	
Pressure Source $P_{21} = P_s$	Volume Flow Source $Q = Q_s$	Continuity $Q_A + Q_B + Q_C = 0$	Compatibility $P_{r1} + P_{12} + P_{2r} = 0$

T → R Transducer



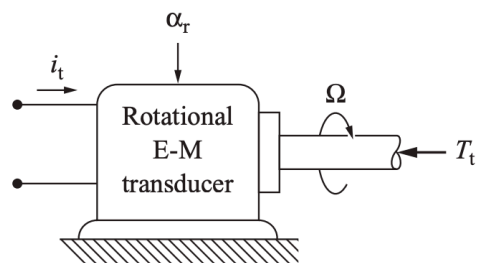
$$\Omega(t) = nV(t) \quad F_t(t) = nT_t(t)$$

E → M Transducer



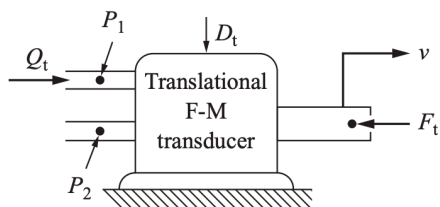
$$v(t) = \alpha_t V_i(t) \quad i_t(t) = \alpha_t F_t(t)$$

E → M Transducer



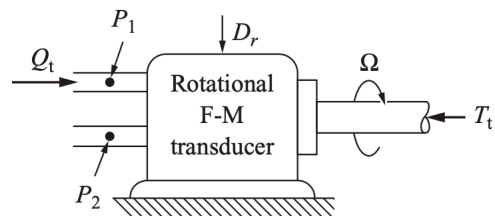
$$\Omega(t) = \alpha_r V_i(t) \quad i_t(t) = \alpha_r T_t(t)$$

F → M Transducer



$$F_t(t) = D_t P_{12}(t) \quad Q_t(t) = D_t V(t)$$

F → M Transducer



$$T(t) = D_r P_{12}(t) \quad Q_t(t) = D_r \Omega(t)$$

Intro to Mechanical Systems

1. Springs

- a. Translational: $F_s = kx(t)$
 - i. The direction of the spring force always opposes positive displacement.
- b. Rotational: $T_s = \mathcal{K}\theta(t)$
 - i. The direction of the torque opposes the displacement of $\theta(t)$
- c. If two masses are connected to one spring, does the force is $k(x_1(t) - x_2(t))$.
Always put the spring force opposite the displacement.

2. Dampers

- a. Translational: $F_d = bv(t) = b \frac{dx(t)}{dt}$
 - i. The direction of the Damper force always opposes positive velocity
- b. Rotational: $T_d = B\Omega(t) = \frac{d\theta(t)}{dt}$ Where B is the damping constant and omega is the angular velocity.
 - i. The torque caused by the damper is opposite of the angular velocity

3. Masses

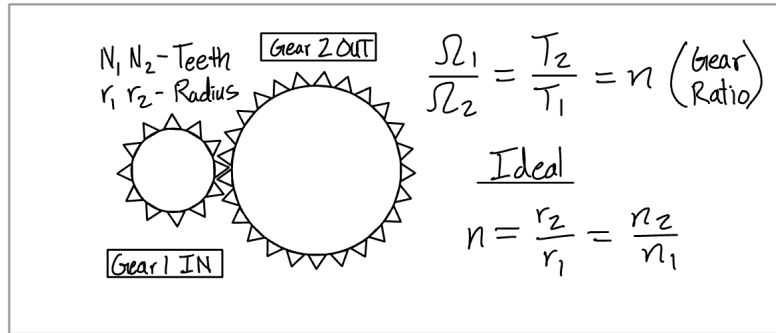
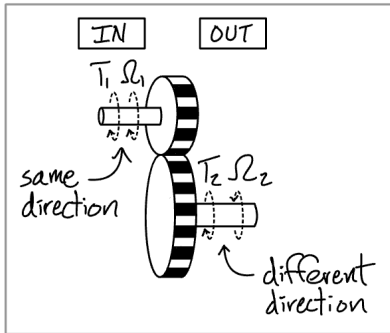
- a. Translational: $\sum F = ma = m \frac{v(t)}{dt} = m \frac{d^2x(t)}{dt^2}$
- b. Rotational: $\sum T = J\alpha = \frac{Jd\Omega(t)}{dt} = J \frac{d^2\theta(t)}{dt^2}$ where J is the mass moment of inertia

4. Gears: Gears are used to amplify angular displacement/velocity/acceleration or torque through a proportional constant.

- a. Input: $\Omega_{out}(t) = \frac{1}{n}\Omega_{in}(t)$, $T_{out}(t) = nT_{in}(t)$
- b. Output: $n = \frac{\Omega_1}{\Omega_2} = \frac{T_2}{T_1}$, $T_{out} = nT_{in}$
- c. Gear Ratio: $n = \frac{n_2}{n_1} = \frac{r_2}{r_1}$, where n is the number of teeth
- d. When doing input, the gear's torque and angular velocity are in the same direction. The output angular velocity is in the opposite direction to the input, but the output torque is in the same direction as the input. The output torque and angular velocity are opposites.

Gears

Used to amplify angular displacement/velocity/acceleration
or torque through a proportional constant.

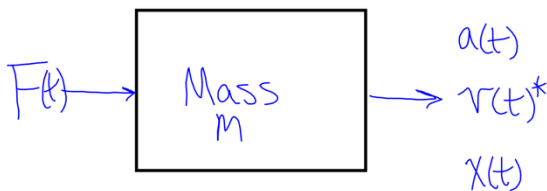


2.2.1 and 2.3.1 Masses

Translational

$$\Sigma F = ma = m \frac{dr(t)}{dt} = m \frac{d^2x(t)}{dt^2}$$

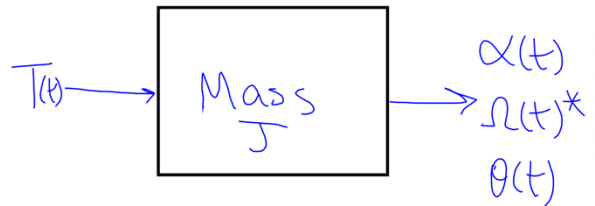
T System



Rotational ^{Mass moment of inertia: J}

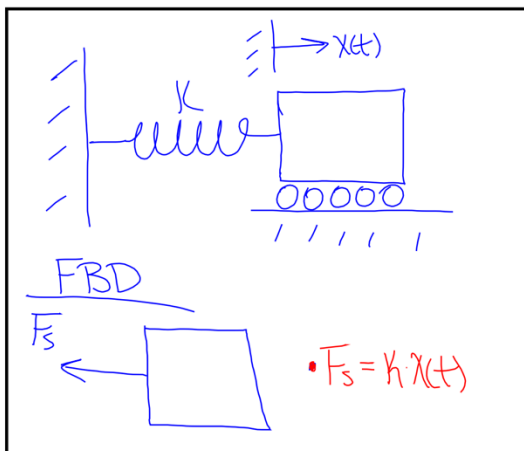
$$\Sigma T = J\alpha = J \frac{d\omega(t)}{dt} = J \frac{d^2\theta(t)}{dt^2}$$

R System

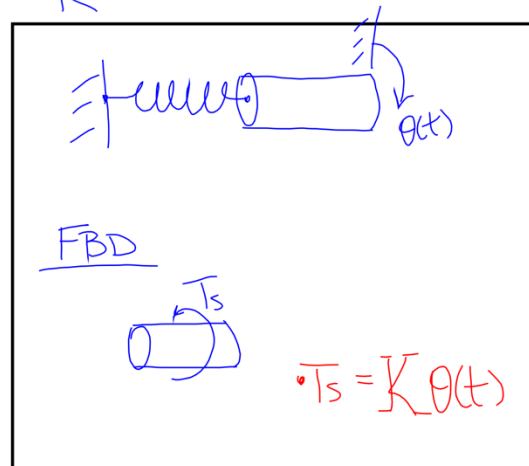


2.2.2 and 2.3.2. Springs

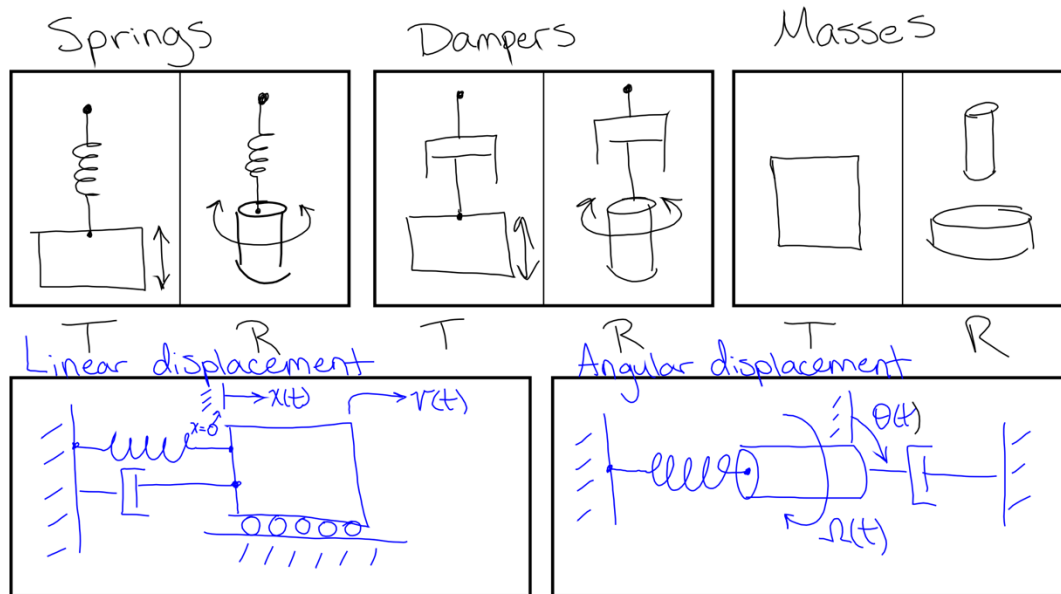
T



R



2.1. Introduction (Mechanical Systems)



Electrical Systems

Capacitor: $i_c(t) = C \frac{dv_c(t)}{dt}$

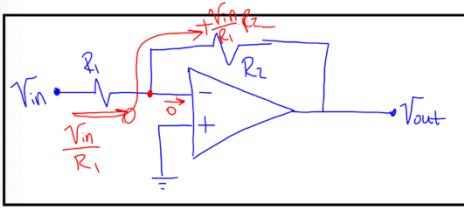
Inductor: $V_l = L \frac{di_l(t)}{dt}$

Resistor: $V(t) = i(t)R$ or $i_R(t) = \frac{1}{R} V_R(t)$

Op Amps

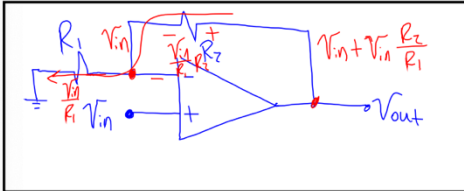
1. $V_1 = V_2$
2. $i_1 = i_2 = 0A$
3. $-V_s \leq V_o \leq V_s$

7.5.1. Inverting Amplifier



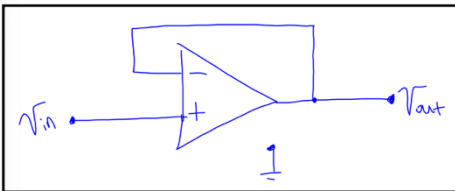
$$V_{out}(t) = -\frac{R_2}{R_1} \cdot V_{in}(t)$$

7.5.2. Noninverting Amplifier



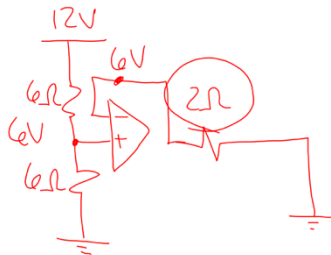
$$V_{out}(t) = \left(1 + \frac{R_2}{R_1}\right) V_{in}(t)$$

7.5.3. Voltage Follower



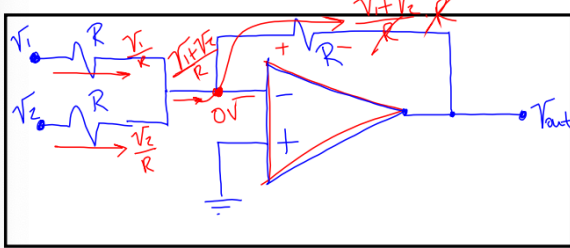
$$V_{out}(t) = V_{in}(t)$$

Why?



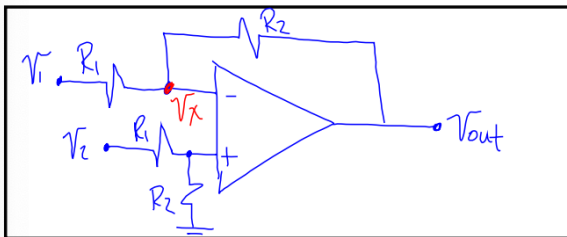
Voltage regulator

7.5.4. Summing Amplifier



$$V_{out}(t) = -(V_1 + V_2)$$

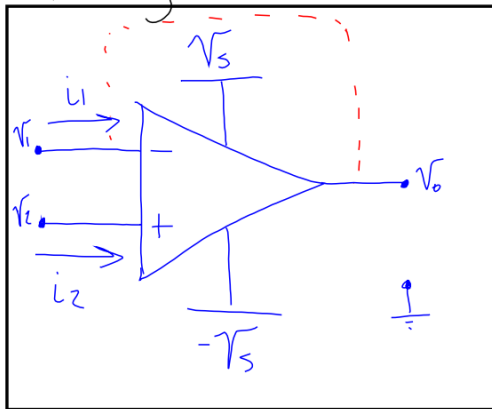
7.5.5. Differential Amplifier



$$V_{out} = \left(\frac{R_2}{R_1} \right) (V_2 - V_1)$$

7.5. Op Amps

Diagram



Characteristics

$$V_1 = V_2$$

$$i_1 = i_2 = 0A$$

$$-V_s \leq V_o \leq V_s$$

"linear"

Fluid Systems

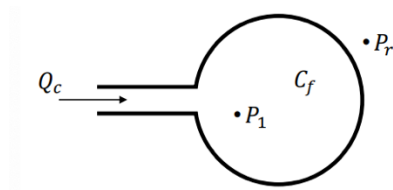
Symbols

- Pressure symbol is used by a dot: •
- Absolute pressure is denoted by P_a
- Gauge is denoted by: $P_{ab} = P_a - P_b$

- Flow Rate or volume flow $\frac{m^3}{s}$

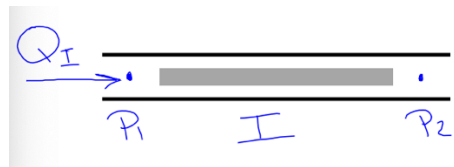
Fluidic System	Electrical System
Q: volume flow rate through	I: current (through)
P: pressure (across)	V: voltage (across)

Fluid capacitors:



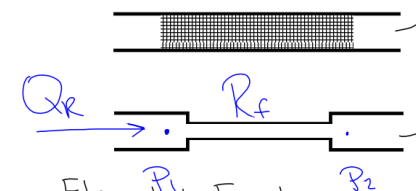
- Flow rate is given by Q_c
- There is an inside pressure and outside pressure. The pressure is the difference between the two. Like voltage.
- $Q_c = \frac{C_f d(P_1 - P_r)}{dt} = C_f \frac{dP_{1f}}{dt}$
- $C_f = \frac{\Delta V}{\Delta P}$

Fluid Inertors



- $P_{12} = I \frac{dQ_I}{dt}$
- $I = \frac{\rho}{A} L$
 - Where ρ is the density of the fluid, A is cross sectional area, L is length.

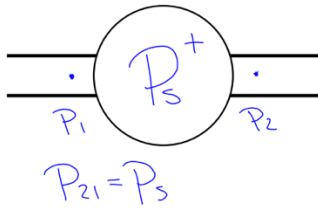
Fluid Resistor



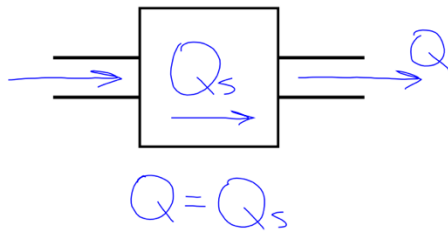
$$P_{12} = R_f Q_R$$

Fluid Sources

- Ideal pressure source

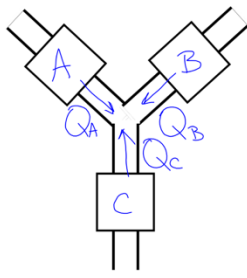


- Ideal flow Source



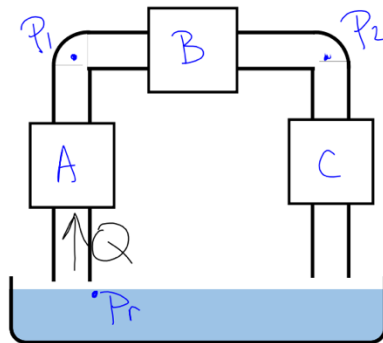
Interconnection Laws

- Continuity law



$$Q_A + Q_B + Q_C = 0$$

- Compatibility law



$$P_{r1} + P_{12} + P_{2r} = 0$$

A-Type variables and elements

System type	Mechanical Translation	Mechanical rotational	Electrical	Fluid
Variable	Velocity, v	Velocity, Ω	Voltage, V	Pressure, P
Element	Mass, m	Mass, J	Capacitor, C	Fluid Capacitor, C_f
Elemental Equation	$F = m \frac{dv}{dt}$	$T = J \frac{d\Omega}{dt}$	$i_c = C \frac{dV_c}{dt}$	$Q_c = C_f \frac{dP}{dt}$
Energy equation	$\epsilon_k = \frac{1}{2}mv^2$	$\epsilon_k = \frac{1}{2}J\Omega^2$	$\epsilon_E = \frac{1}{2}CV_c^2$	$\epsilon_P = \frac{1}{2}C_fP^2$

T-Type variables and elements

System type	Mechanical Translation	Mechanic rotational	Electrical	Fluid
Variable	Force, F	Torque, T	Current, i	Volumetric flow rate, Q_f
Element	Compliance, $\frac{1}{k}$	Compliance, $\frac{1}{\mathcal{K}}$	Inductor, L	Inertor, I
Elemental Equation	$V = \frac{1}{k} \cdot \frac{dF}{dt}$	$\Omega = \frac{1}{\mathcal{K}} \cdot \frac{dT}{dt}$	$V = L \frac{di}{dt}$	$P = I \frac{dQ}{dt}$
Energy equation	$\epsilon_P = \frac{1}{2} \cdot \frac{1}{k} \cdot F^2$	$\epsilon_P = \frac{1}{2} \cdot \frac{1}{\mathcal{K}} \cdot T^2$	$\epsilon_m = \frac{1}{2}Li^2$	$\epsilon_k = \frac{1}{2}IQ^2$

D-Type elements

System Type	Mechanical Translational	Mechanical rotational	Electrical	Fluid
Element	Damper, b	Damper, B	Resistor, R	Fluid Resistor, R_f
Elemental Equation	$F = bv$	$T = B\Omega$	$i = \frac{1}{R} \cdot V$	$Q_f = \frac{1}{R_f} P$

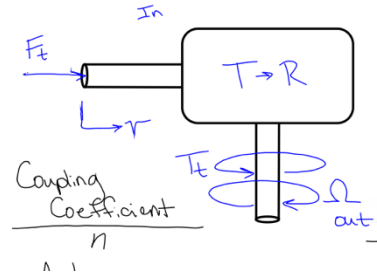
Mixed Systems

Transducers and Devices

- Transducer: Devices that have been developed to connect different types of systems.
- Ideal Energy – Converting Transducers: Transducers that do not store or dissipate energy. All energy from one system is transferred to another.
- Translational – Mechanical to Rotational - Mechanical
 - A Type: $\Omega(t) = nV(t)$
 - T Type: $nT_t(t) = F_t(t)$

- A and T-type variables are proportional to one another across system types.

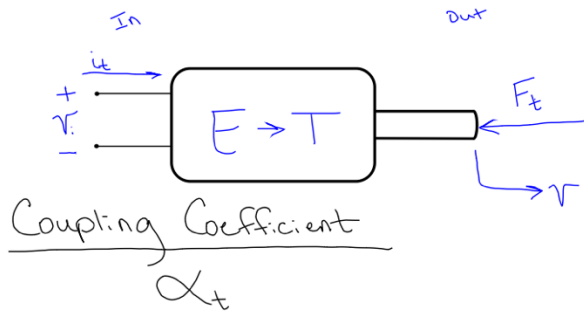
$$VF_t = \Omega(t)T_t(t) = \Omega T_t$$



- Electromechanical energy converters

- A Type: $V(t) = \alpha_t V_i(t)$
- T Type: $\alpha_t F_t(t) = i_t(t)$

$$V_i(t)i_t(t) = V(t)F_t(t)$$



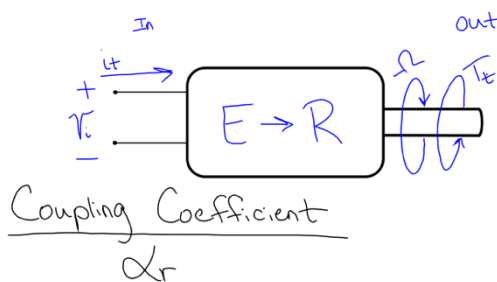
Linear Actuator



- Electromechanical energy converters

- A Type: $\Omega(t) = \alpha_r V_i(t)$
- T Type: $\alpha_t T_t(t) = i_t(t)$

$$i_t(t)V_i(t) = \Omega(t)T_t(t)$$

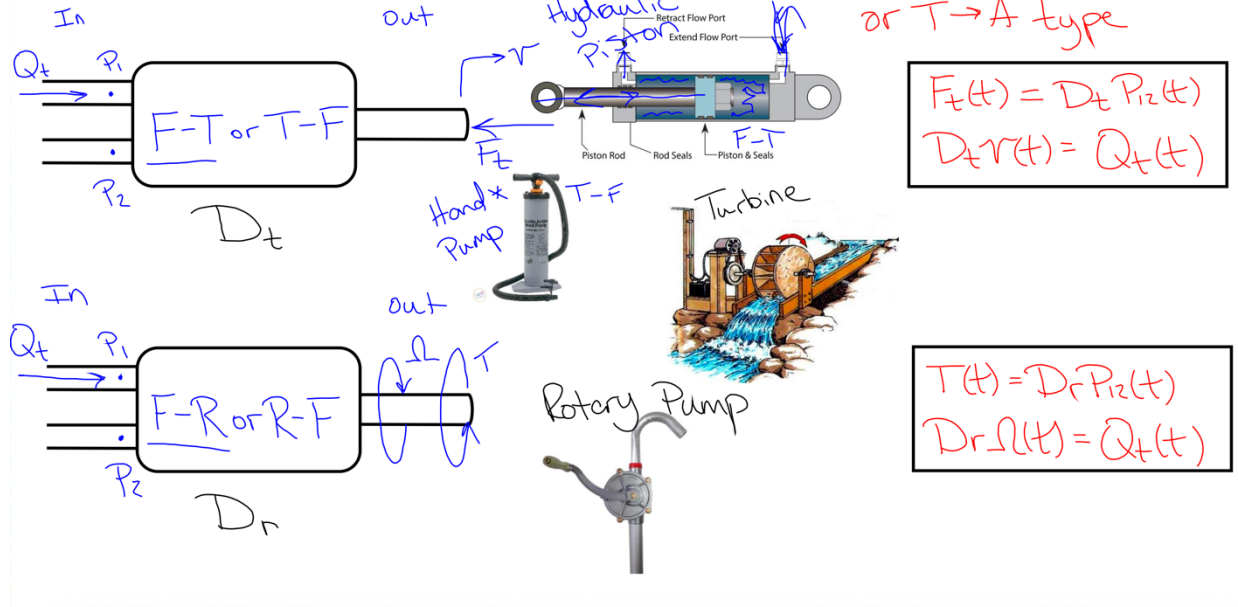


DC Motor



Fluid Mechanical energy converters

10.2.3. Fluid Mechanical energy converters



- F-T or T-F
 - $F_t(t) = D_t P_{12}(t)$
 - $D_t v(t) = Q_t(t)$
- F-R or R-F
 - $T(t) = D_r P_{12}(t)$
 - $D_r \Omega(t) = Q_t(t)$
- Other Transducers
 - Stirling Engine TH-R (Very inefficient)
 - Sensor Transducers (do not convert energy)

Transfer Functions

- Laplace of the nth derivative
 - $\mathcal{L}\left\{\frac{d^n f(t)}{dt^n}\right\} = s^n F(s) - \sum_{k=0}^{n-1} s^{n-k-1} \left[\frac{d^k f(t)}{dt^k}\right]_{t=0}$
 - Assum IC's = 0, $\mathcal{L}\left\{\frac{d^n f(t)}{dt^n}\right\} = s^n F(s)$

Single input single output $H(s) = \frac{F(s)}{x(s)} = \frac{OUTPUT}{INPUT}$

General second-order transfer function:

$$\ddot{y}(t) + d\dot{y}(t) + ey(t) = cx(t) \rightarrow H(s) = \frac{c}{s^2 + ds + e}$$
$$= k \frac{\omega_0^2}{s^2 + 2\zeta\omega_0s + \omega_0^2}$$

Where

- System gain $k = \frac{c}{e}$
- Damping Ratio $\zeta = \frac{d}{2\sqrt{e}}$
- Natural Frequency (no damping) $\omega_0 = \sqrt{e}$

Properties of transfer function

$$H(s) = \frac{B(s)}{A(s)}$$

- Polynomial Num & Den

$$B(s) = bms^m + \dots + b_1s + b_0$$

$$A(s) = s^n + a_{n-a}s^{s-1} + \dots + a_1s + a_0$$

$$n \geq m$$

- Zeros – Numerator $H(s) = 0$

$$B(s) = mb(s - z_1)(s - z_2) \dots (s - z_m)$$

- Poles – Denominator $H(s) = \infty$

$$A(s) = (s - p_1)(s - p_2) \dots (s - p_n)$$

General 2nd order

$$H(s) = k \frac{\omega_0^2}{s^2 + 2\zeta\omega_0s + \omega_0^2}$$

Solving for poles, we get

$$P_{1,2} = \sigma \pm j\omega$$

Where:

- σ is decay
- ω is frequency

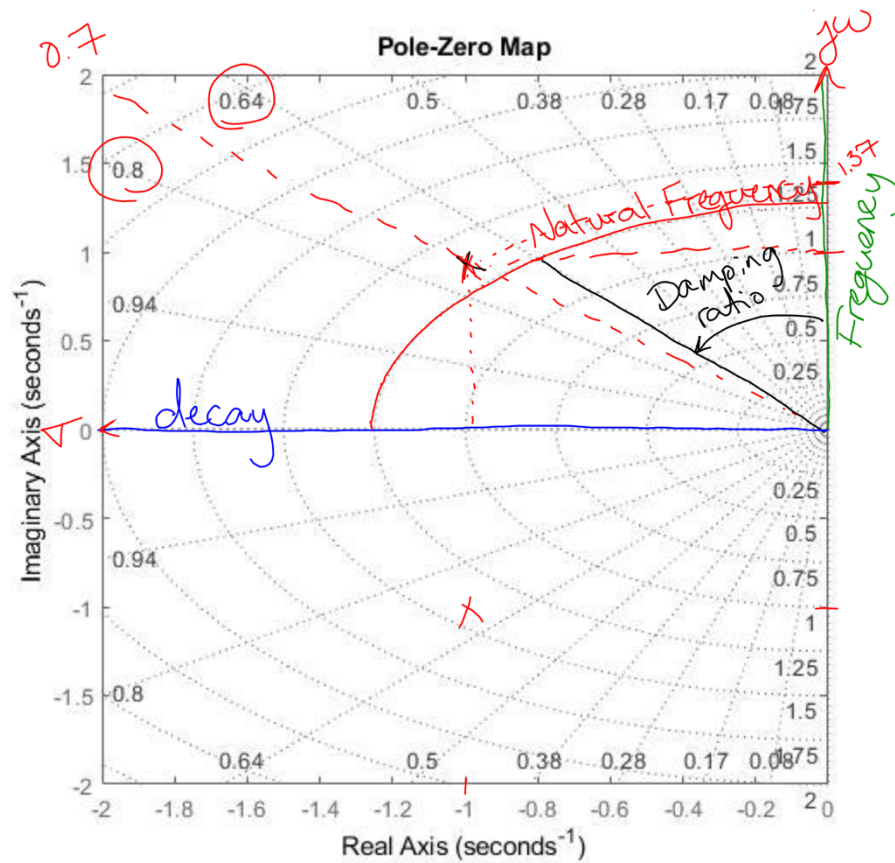
Poles tell us

1. Damping ratio (under or over)
2. Natural frequency (undamped frequency) - ω_0
3. Response frequency (damped frequency) - ω
4. Decay rate (stability) - σ
5. gain

Pole-zero map

- Closer to the imag axis, the slower the dissipation (dominance)
- The distance from the center to the pole is the natural frequency.
- The distance from the real axis or farther up is the damped frequency
- The damping ratio is directly related to the angle. Distance from origin are lines of constant damping ratio

Step Response for Underdamped



$$P_{1,2} = -1 \pm j1$$

Step Response for Overdamped

Step Response for Final Value

$$y(\infty) = \lim_{s \rightarrow 0} [A \cdot H(s)]$$

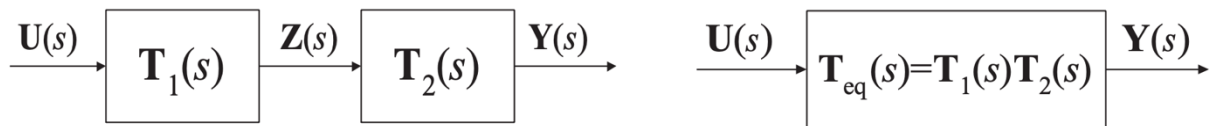
Where:

- A is the constant value of the driving force

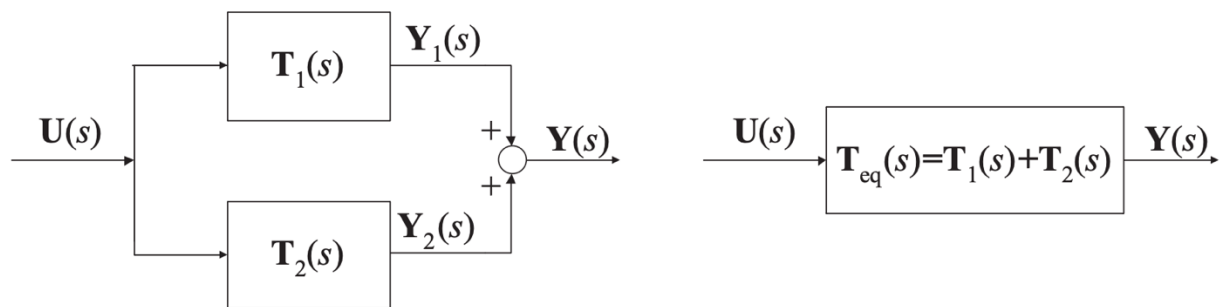
Block diagram algebra

Combining blocks

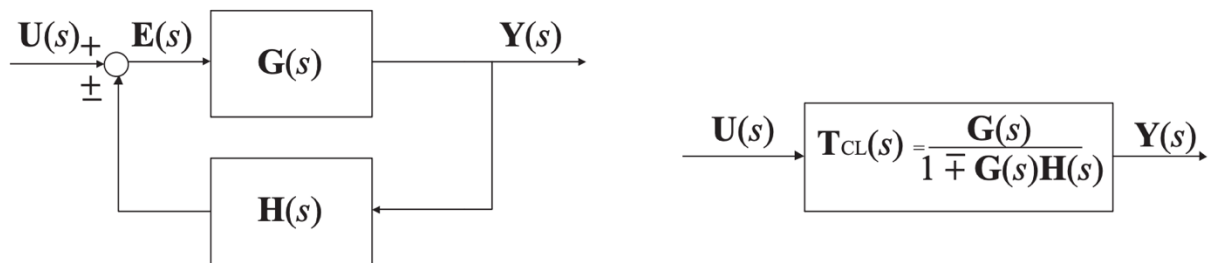
Series:



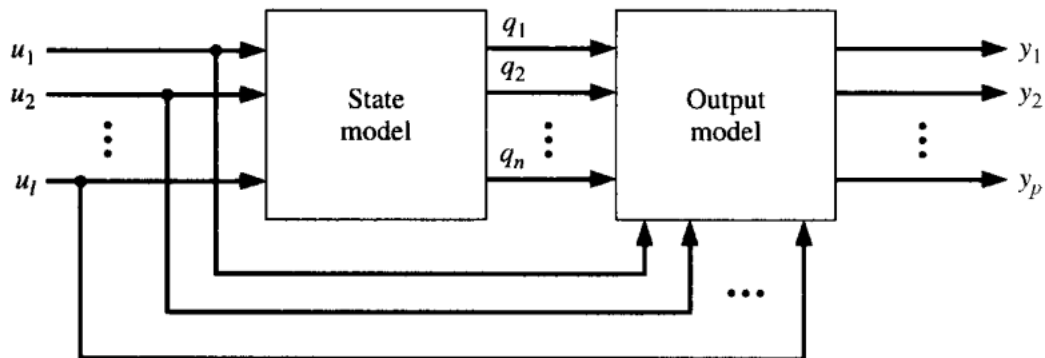
Parallel:



Feedback:



State Space (time domain): a model that describes the relationship between the inputs, outputs, parameters, and states.



State Model

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \vdots \\ \dot{q}_n \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1l} \\ b_{21} & b_{22} & \cdots & b_{2l} \\ \vdots & \vdots & & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nl} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_l \end{bmatrix},$$

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & & \vdots \\ c_{p1} & c_{p2} & \cdots & c_{pn} \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix} + \begin{bmatrix} d_{11} & d_{12} & \cdots & d_{1l} \\ d_{21} & d_{22} & \cdots & d_{2l} \\ \vdots & \vdots & & \vdots \\ d_{p1} & d_{p2} & \cdots & d_{pl} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_l \end{bmatrix}$$

$$\vec{\dot{q}} = \vec{A}\vec{q} + \vec{B}\vec{u}$$

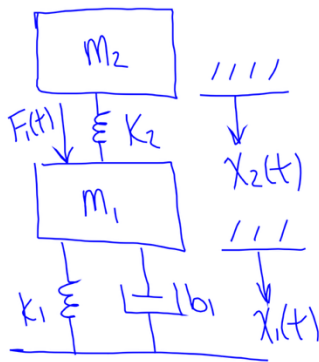
$$\vec{y} = \vec{C}\vec{q} + \vec{D}\vec{u}$$

If nonlinear

$$\vec{\dot{q}} = \vec{A}\vec{q} + \vec{\rho} + \vec{B}\vec{u}$$

Determining the States of a System

1. The Output and its derivatives: when your model can be described as a single output with higher-order derivatives
 - a. $q_1(t) = x(t)$
 - b. $q_2(t) = \dot{x}(t) = \dot{q}_1(t)$
2. T-type and A-type variables
 - a. Example A-type $\rightarrow$ mass(velocity), T-type spring(force)
 - b. $q_1(t) = F_{k1}(t) = k_1 x_1(t)$
 - c. $q_2(t) = F_{k2}(t) = k_2(x_2(t) - x_1(t))$
 - d. $q_3(t) = \dot{x}_1(t)$
 - e. $q_4(t) = \dot{x}_2(t)$



1. Two or more variables and their derivatives
 - a. If you have two masses with different outputs. m_1 has a position and velocity and m_2 has a position and velocity. Could be angles as well.
 - b. $q_1(t) = x_1(t)$
 - c. $q_2(t) = \dot{x}_1(t)$
 - d. $q_3(t) = x_2(t)$
 - e. $q_4(t) = \dot{x}_2(t)$

Simulations

Simulation: A numerical solution (and often, graphical representation) to differential equations that describe physical systems.

Euler's Method

$$\dot{y}(t) = f(y(t), t)$$

Approximation

$$\frac{dy(t)}{dt} \approx \frac{y(t + \Delta t) - y(t)}{\Delta t}$$

Predict $y(t + \Delta t)$

$$y(t + \Delta t) = y(t) + \Delta t \cdot f(y(t), t)$$

Multiple States and/or higher order systems

$$\vec{q}(t) = \vec{A}\vec{q} + \vec{p} + \vec{B}u$$

$$\vec{f}(\vec{q}(t), t) = \vec{A}\vec{q}(t) + \vec{p}(t) + \vec{B}u(t)$$

$$\vec{q}(t + \Delta t) = \vec{q}(t) + \Delta t \left(\vec{A}\vec{q}(t) + \vec{p}(t) + \vec{B}u(t) \right)$$

Improved Eulers Method

Put chart here

Approximation of the slope t

$$\frac{dy(t)}{dt} = f(y(t), t) = k_1$$

Euler's for int. value

$$\hat{y}(t + \Delta t) = y(t) + \Delta t \cdot k_1$$

Approximation of the slope at $t + \Delta t$

$$f(\hat{y}(t + \Delta t), t + \Delta t) = k_2$$

Average the slopes

$$k_3 = \frac{k_1 + k_2}{2}$$

Estimate our true prediction with k_3

$$y(t + \Delta t) = y(t) + \Delta t \cdot k_3$$

Runge Kutta Method

$$k_1 = f(x(t), t)$$

$$\hat{x}_1 = x(t) + k_1 \cdot \frac{\Delta t}{2}$$

$$k_2 = f\left(\hat{x}_1, t + \frac{\Delta t}{2}\right)$$

$$\hat{x}_2 = x(t) + k_2 \cdot \frac{\Delta t}{2}$$

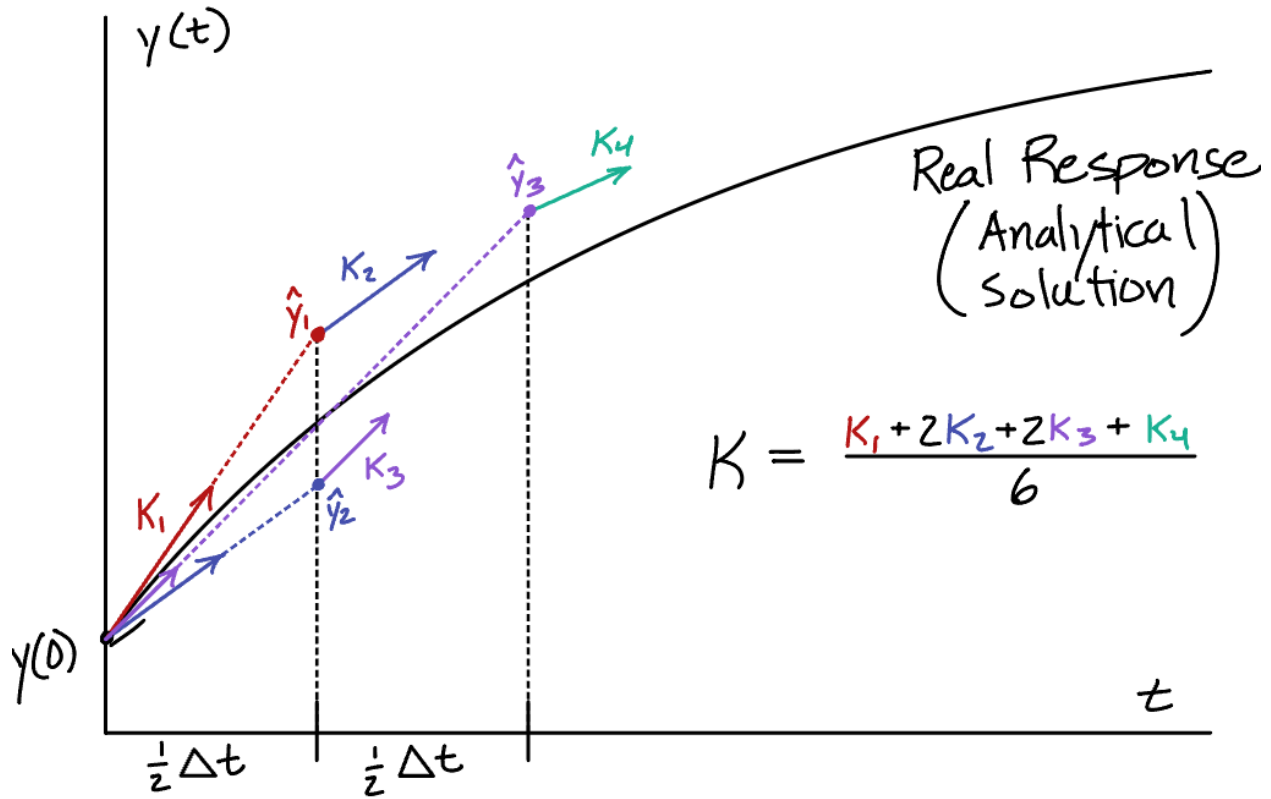
$$k_3 = f\left(\hat{x}_2, t + \frac{\Delta t}{2}\right)$$

$$\hat{x}_3 = x(t) + k_3 \cdot \Delta t$$

$$k_4 = f(\hat{x}_3, t + \Delta t)$$

$$k = \frac{k_1 + 2k_2 + 2k_3 + k_4}{6}$$

$$x(t + \Delta t) = x(t) + \Delta t \left(\frac{k_1 + 2k_2 + 2k_3 + k_4}{6} \right)$$



$$k_2 = f\left(\hat{x}_1\left(t + \frac{\Delta t}{2}\right), t + \frac{\Delta t}{2}\right) = f\left(x(t) + k_1 \cdot \frac{\Delta t}{2}, t + \frac{\Delta t}{2}\right)$$

$$k_3 = f\left(\hat{x}_2\left(t + \frac{\Delta t}{2}\right), t + \frac{\Delta t}{2}\right) = f\left(x(t) + k_2 \cdot \frac{\Delta t}{2}, t + \frac{\Delta t}{2}\right)$$

$$k_4 = f(\hat{x}_3(t + \Delta t), t + \Delta t) = f(x(t) + k_3 \cdot \Delta t, t + \Delta t)$$

The true prediction is

$$k = \frac{k_1 + 2k_2 + 2k_3 + k_4}{6}$$

$$x(t + \Delta t) = x(t) + \Delta t \left(\frac{k_1 + 2k_2 + 2k_3 + k_4}{6} \right)$$

What is a digital system?

Continuous systems

- Function with infinity many point between and two points.
Physical world $y(t)$, $x(t)$, $q(t)$
- Differential Equations & Laplace Transforms

Digital systems

- Digital (discrete system)
- Depends on bit resolution
- Z Transform

Discretize function

Discretizing Differential Equations

$$a_1 \cdot \frac{dy(t)}{dt} + a_0 y(t) = b_1 \cdot \frac{x(t)}{dt} + b_0 x(t)$$

T_s sampling period

$$\frac{dy(t)}{dt} \approx \frac{y(t + T_s) - y(t)}{T_s} \rightarrow \frac{y(t) - y(t - T_s)}{T_s}$$

These replace terms in the differential equation

$$y(t) \rightarrow y(nT_s) \rightarrow y(n)$$

$$y(t - T_s) \rightarrow y(nT_s - T_s) \rightarrow y(n - 1)$$

$$\frac{dy(t)}{dt} \rightarrow \frac{y(n) - y(n-1)}{T_s}$$

For open form replace t with nT_s

$$\text{EX: } 2 \cos(2\pi 1t) \rightarrow 2 \cos(2\pi 1nT_s)$$

For closed form plug in the number of given n values

$$\text{EX: } n = 0, 1, 2, 3$$

$$2 \cos(2\pi 1t)$$

$$\rightarrow 2 \cos(2\pi T_s 0), 2 \cos(2\pi T_s 1), 2 \cos(2\pi T_s 2), 2 \cos(2\pi T_s 3)$$

Difference equation

$$-\frac{a_1}{T_s} y(n-1) + \left(\frac{a_1}{T_s} + a_0\right) y(n) = -\frac{b_1}{T_s} x(n-1) + \left(\frac{b_1}{T_s} + b_0\right) x(n)$$

Difference Equations (Input-Output Models)

$$y(n) = \frac{-b_1}{T_s \left(\frac{a_1}{T_s} + a_0\right)} x(n-1) + \frac{\left(\frac{b_1}{T_s} + b_0\right)}{\left(\frac{a_1}{T_s} + a_0\right)} x(n) + \frac{a_1}{T_s \left(\frac{a_1}{T_s} + a_0\right)} y(n-1)$$

Discretizing Second (and more) Derivatives

$$\frac{d^2 y(t)}{dt^2} \approx \frac{\frac{dy(t)}{dt} - \frac{dy(t-T_s)}{dt}}{T_s} \approx \frac{y(n) - 2y(n-1) + y(n-2)}{T_s^2}$$

Digital Filters

We can evaluate the frequency response of a system by plugging in $s = j\omega$

Example:

$$F(s) = \frac{k}{s + k}$$

$$F(j\omega) = \frac{k}{j\omega + k} \rightarrow (\text{low pass filter}, F(j0) = 1, F(j\infty) = 0)$$

The amplitude at which the cutoff frequency occurs is

$$|F(j\omega_c)| = \frac{\text{maxmag}}{\sqrt{2}} = \frac{1}{\sqrt{2}} = \frac{k}{\sqrt{\omega_c^2 + k^2}}$$

$$\left(\sqrt{\omega_c^2 + k^2} = k\sqrt{2} \right)^2 \rightarrow \omega_c = \pm k = k$$

The laplace is:

$$F(s) = \frac{Y(s)}{X(s)} = \frac{\omega_c}{s + \omega_c}$$

Inverse laplace:

$$\dot{y}(t) + \omega_c \cdot y(t) = \omega_c x(t)$$

Disc

$$y(n) = \frac{1}{1 + T_s \omega_c} [y(n-1) + T_s \omega_c x(n)]$$

High-pass filter

$$F(s) = \frac{s}{s + \omega_c}$$

$$F(j0) = 0 \quad F(j\infty) = 1$$

$$\frac{Y(s)}{X(s)} = \frac{s}{s + \omega_c} \rightarrow Y(s) + \omega_c Y(s) = sX(s)$$

Inverse laplace

$$\dot{y}(t) + \omega_c \cdot y(t) = \dot{x}(t)$$

Disc:

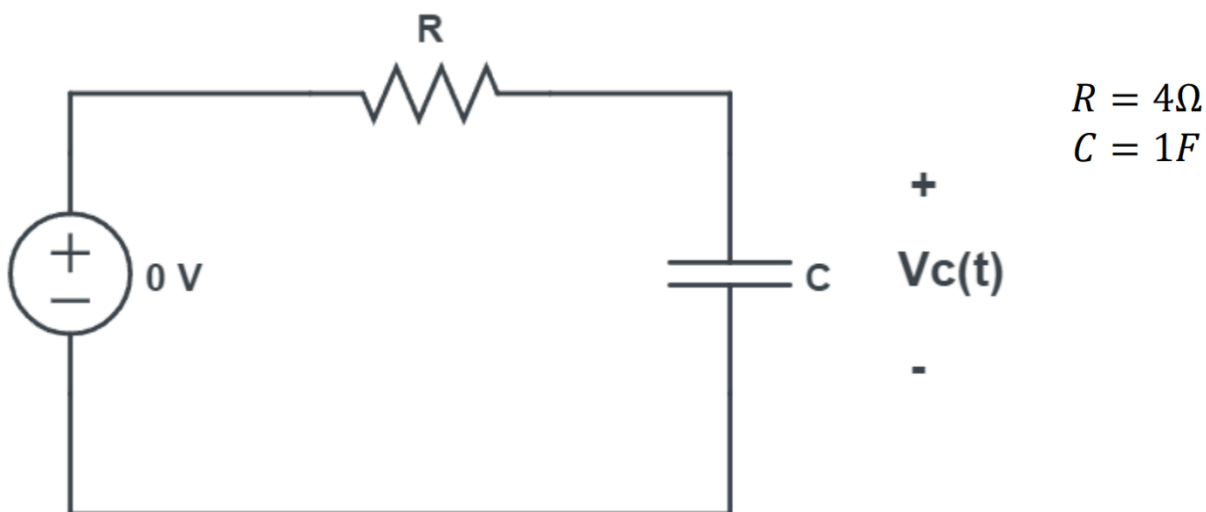
$$y(n) = \left(\frac{1}{\frac{1}{T_s} + \omega_c} \right) \left(\frac{1}{T_s} y(n-1) + \frac{x(n) - x(n-1)}{T_s} \right)$$

Black and Gray boxes

Black box: Order? Linearity? Time-variance? You don't know much at all

Gray box: We know Order? Linearity? Time-variance? We do not know constants

Example:



$$\text{KVL: } -0 + RC\dot{V}_c(t) + V_c(t) = 0 \quad R = 4\Omega, C = 1F$$

$$\dot{V}_c(t) + \frac{1}{RC} \cdot V_c(t) = 0$$

Approach to solve

2. Solve for highest derivative

$$\dot{V}_c(t) = -\frac{1}{RC} V_c(t)$$

3. Lump all unknown coefficients

$$\dot{V}_c(t) = KV_c(t) \text{ where } K = -\frac{1}{RC} \rightarrow \left(-\frac{1}{4}\right)$$

4. We need to discretize

$$\frac{V_c(n+1) - V_c(n)}{T_s} = KV_c(n)$$

$$V_c(n+1) = V_c(n) + T_s KV_c(n) = V_c(n+1, K)$$

An estimation of $V_c(n+1)$ given an n and a random K

Optimization theory function: A function that we want to minimize

Given: N observed points

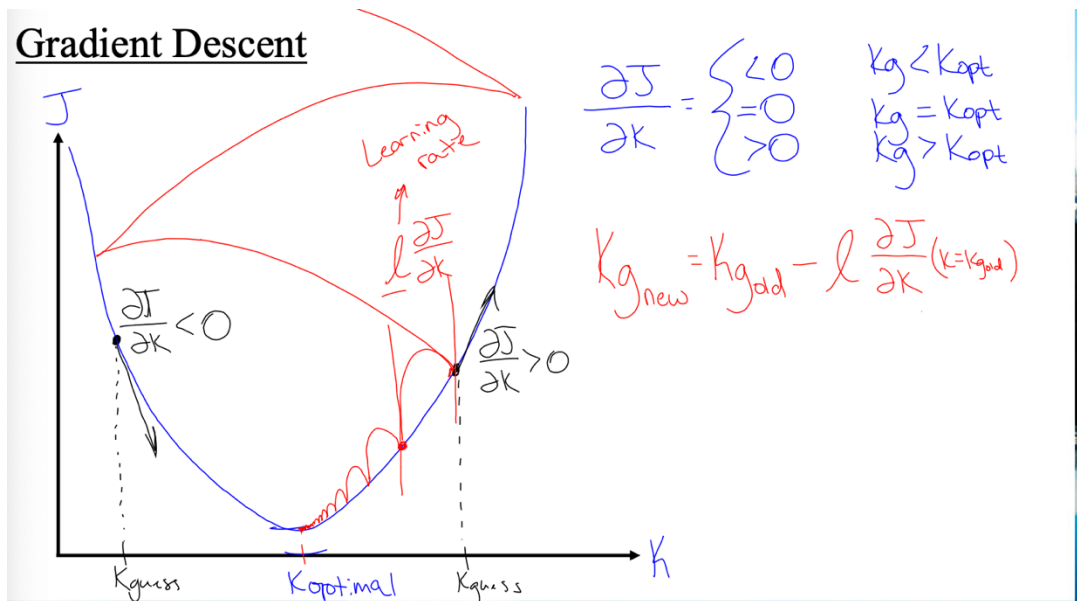
Least Squares Error

$$J = \sum_{n=0}^{N-2} \frac{1}{2} (\text{Estimate} - \text{true})^2$$

$$J = \sum_{n=0}^{N-2} \frac{1}{2} (V_c(n) + T_s KV_c(n) - V_c(n+1))^2$$

Gradient Descent

Gradient Descent



$$J = \sum_{n=0}^{N-2} \frac{1}{2} (V_c(n) + T_s K V_c(n) - V_c(n+1))^2$$

$$\frac{\partial J}{\partial K} = \sum_{n=0}^{N-2} (V_c(n) + T_s K V_c(n) - V_c(n+1)) (T_s V_c(n))$$

Example 2: A cup of coffee is being heated by a heating element. The coffee begins at an initial temperature of 40°C while the heating element transfers $P(t)$ W into the coffee. Identify this system given simulation data.

Convection/Conduction equation

$$Q_{out}(t) = \frac{1}{R} (T(t) - T_A(t))$$

The differential equation

$$\dot{T}(t) = -\frac{1}{RC_h} T(t) + \frac{1}{RC_h} T_A(t) + \frac{\alpha h}{C_h} P(t)$$

We arrive at:

$$\dot{T}(t) = K_1 T(t) + K_3 T_A(t) + K_2 P(t)$$

Discretize

$$T(n+1) = T(n) + T_s K_1 T(n) + T_s K_2 P(n) + T_s K_3 T_A(n)$$

$$\vec{K} = [K_1 \quad K_2 \quad K_3] \quad \vec{S}(n) = \begin{bmatrix} T(n) \\ P(n) \\ T_A(n) \end{bmatrix}$$

$$T(n+1) = T(n) + T_s [K_1 \quad K_2 \quad K_3] \cdot \begin{bmatrix} T(n) \\ P(n) \\ T_A(n) \end{bmatrix}$$

$$T(n+1, \vec{K}) = T(n) + T_s \vec{K} \vec{S}(n)$$

Now we need to use the cost function to find $\vec{K}$

$$J = \sum_{n=0}^{N-2} \frac{1}{2} \{T(n) + T_s \vec{K} \vec{S}(n) - T(n+1)\}^2$$

Taking the partial derivative: to take the partial derivative $\frac{\partial(\vec{A}\vec{X})}{\partial\vec{X}} = \vec{A}^T$

$$\frac{\partial J}{\partial \vec{K}} = \sum_{n=0}^{N-2} \{T(n) + T_s \vec{K} \vec{S}(n) - T(n+1)\} T_s \vec{S}^T(n) = \begin{bmatrix} \frac{\partial J}{\partial K_1} & \frac{\partial J}{\partial K_2} & \frac{\partial J}{\partial K_3} \end{bmatrix}$$

$$\vec{K}_{new} = \vec{K}_{old} - l \frac{\partial J}{\partial \vec{K}}$$

Closed Form Solution

Closed Form Solution

Redefine

$$\vec{T}_n = [T(0), T(1), \dots, T(N-2)]^{1 \times N-1} \quad \vec{T}_{n+1} = [T(1), T(2), \dots, T(N-1)]^{1 \times N-1}$$

$$\vec{S} = [\vec{S}(0), \vec{S}(1), \dots, \vec{S}(N-2)] = \begin{bmatrix} T(0) & T(1) & \dots \\ P(0) & P(1) & \dots \\ I_A(0) & I_A(1) & \dots \end{bmatrix}^{3 \times N-1}$$

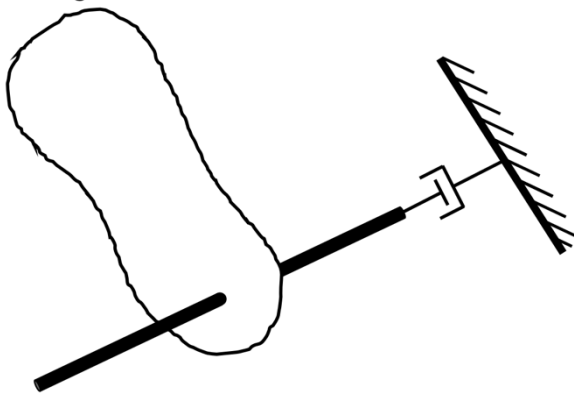
$$\Rightarrow \frac{\partial J}{\partial \vec{K}} = \left\{ \vec{T}_n + T_s \cdot \vec{K} \vec{S} - \vec{T}_{n+1} \right\} T_s \vec{S}^T = [0, 0, 0] = \vec{0}_{1 \times 3}$$

$$= \vec{T}_n \vec{S}^T + T_s \vec{K} \vec{S} \vec{S}^T - \vec{T}_{n+1} \vec{S}^T = \vec{0}$$

$$T_s \vec{K} \vec{S} \vec{S}^T = (\vec{T}_{n+1} - \vec{T}_n) \vec{S}^T \quad \boxed{\vec{K} = \frac{1}{T_s} (\vec{T}_{n+1} - \vec{T}_n) \vec{S}^T (\vec{S} \vec{S}^T)^{-1}}$$

Example 5.3 – Swinging Free-form

A mass of arbitrary shape is attached to a bar that is free to swing that has some damping in the bearings. Identify this system using the close-form solution to a cost function.



- Mass ($M = 1\text{ kg}$)
- Distance to COM ($L = 0.25\text{ m}$)
- Moment of inertia ($J = 1.2 \frac{\text{Nms}^2}{\text{rad}}$)
- Rotational Damping ($B = 0.2 \frac{\text{Nms}}{\text{rad}}$)

$$\Sigma T_\theta = -B\dot{\theta}(t) + MgL\sin(\theta(t)) = J\ddot{\theta}(t)$$

Step one: highest order

$$\ddot{\theta}(t) = \frac{MgL}{J} \sin(\theta(t)) = \frac{B}{J} \dot{\theta}(t)$$

$$\ddot{\theta}(t) = K_1 \sin(\theta(t)) + K_2 \dot{\theta}(t)$$

$$K_1 = \frac{MgL}{J} \approx 2.04 \quad K_2 = -\frac{B}{J} \approx -0.167$$

Those K values are what we hope to find.

$$\ddot{\theta}(t) = K_1 \sin(\theta(t)) + K_2 \dot{\theta}(t)$$

On the left side, we need exactly 1 n+1 term

We can approximate with

$$\frac{\theta(n+1) - 2\theta(n) + \theta(n-1)}{T_s^2}$$

$$\frac{\theta(n+1) - 2\theta(n) + \theta(n-1)}{T_s^2} = K_1 \sin(\theta(t)) + K_2 \cdot \frac{\theta(n) - \theta(n-1)}{T_s}$$

Solve for $\theta(n+1)$

$$\theta(n+1, \vec{K}) = 2\theta(n) - \theta(n-1) + T_s^2 [K_1 \quad K_2] \left| \begin{array}{c} \sin(\theta(n)) \\ \frac{\theta(n) - \theta(n-1)}{T_s} \end{array} \right|$$

$$\theta(n+1, \vec{K}) = 2\theta(n) - \theta(n-1) + T_s^2 \vec{K} \vec{s}(n)$$

$$J = \sum_{n=1}^{N-2} \frac{1}{2} \{ 2\theta(n) - \theta(n-1) + T_s^2 \vec{K} \vec{s}(n) - \theta(n+1) \}^2$$

$$\frac{\partial J}{\partial \vec{K}} = \sum_{n=1}^{N-2} \{ 2\theta(n) - \theta(n-1) + T_s^2 \vec{K} \vec{s}(n) - \theta(n+1) \} \vec{s}^T(n)$$

Constant at the end was removed

Solve with closed form, we need to make vectors

$$\vec{\theta}_n = [\theta(1) \quad \theta(2) \quad \dots \quad \theta(N-2)]_{1 \times N-2}$$

$$\vec{\theta}_{n-1} = [\theta(0) \quad \theta(1) \quad \dots \quad \theta(N-3)]_{1 \times N-2}$$

$$\vec{\theta}_{n+1} = [\theta(2) \quad \theta(3) \quad \dots \quad \theta(N-1)]_{1 \times N-2}$$

$$\vec{s}_n = [\vec{s}(1) \quad \vec{s}(2) \quad \dots \quad \vec{s}(N-2)] = \begin{bmatrix} \sin(\vec{\theta}_n) \\ \frac{\vec{\theta}_n - \vec{\theta}_{n-1}}{T_s} \end{bmatrix}_{\rightarrow 2 \times N-2}$$

$$\frac{\partial J}{\partial \vec{K}} = (2\vec{\theta}_n - \vec{\theta}_{n-1} + T_s^2 \vec{K} \vec{s}_n - \theta_{n+1}) \vec{s}_n^T = [0 \quad 0] = \vec{0}_{1 \times 2}$$

$$\frac{\partial J}{\partial \vec{K}} = 2\vec{\theta}_n \vec{s}_n^T - \vec{\theta}_{n-1} \vec{s}_n^T + T_s^2 \vec{K} \vec{s}_n \vec{s}_n^T - \theta_{n+1} \vec{s}_n^T = 0$$

$$T_s^2 \vec{K} (\vec{s}_n \vec{s}_n^T)_{2 \times 2} = (-2\vec{\theta}_n + \vec{\theta}_{n-1} + \vec{\theta}_{n+1}) \vec{s}_n^T$$

$$\vec{K} = \frac{1}{T_s^2} (-2\vec{\theta}_n + \vec{\theta}_{n-1} + \vec{\theta}_{n+1}) \vec{s}_n^T (\vec{s}_n \vec{s}_n^T)^{-1}$$

